BMP Extension - Complete Technical Documentation

> **Status:** COMPLETED
> **Last Updated:** March 2026
> **Project:** Activepieces BMP Multi-Tenant Extension

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Executive Summary

The BMP (Business Process Management) extension adds multi-tenant organization support to Activepieces Community Edition. The implementation follows a **hook-based architecture** that enables conditional loading of BMP features without modifying core Activepieces functionality.

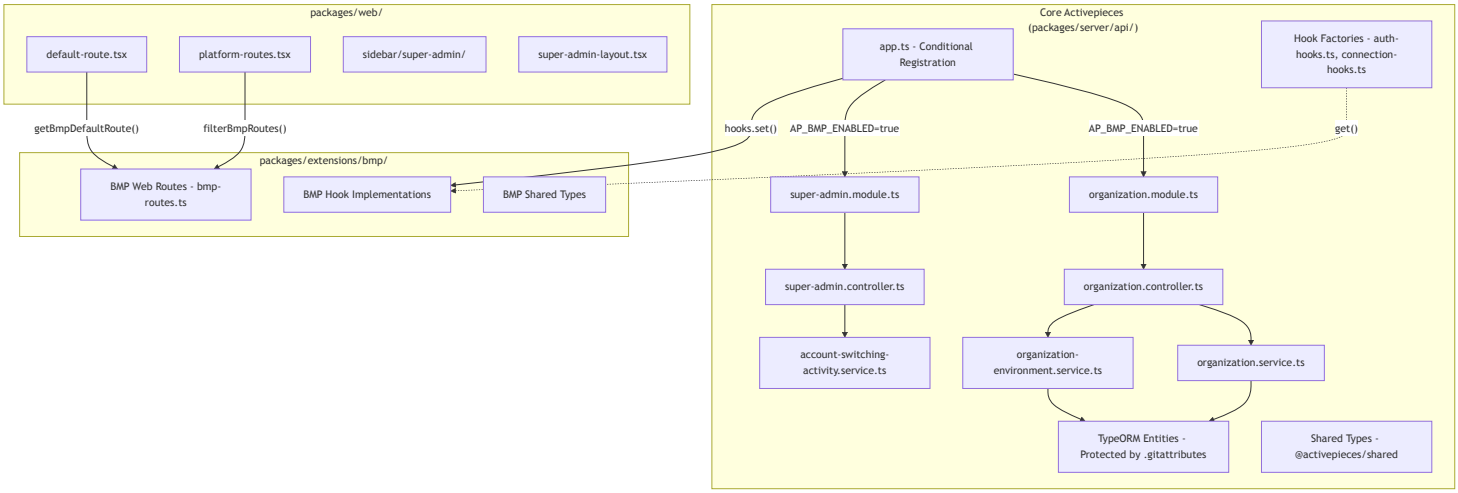
Key Statistics

Category	Count
New BMP files	86 (including 40 in react-ui-sdk)
Core files modified	18
New core hook files created	3
Extension package files	12
Features validated	25+
Total files changed	37 (excluding new BMP files)

What Was Built

1. **Multi-tenant Organization System** - Organizations with environments (Dev/Staging/Production)
2. **Extended User Roles** - Added `SUPER_ADMIN` and `OWNER` to existing hierarchy
3. **BMP Piece Integration** - Custom ada-bmp piece with environment-aware API URL
4. **Account Switching** - Stack-based session management for cross-role navigation
5. **UI Customizations** - New routes, sidebars, dialogs for organization management
6. **React UI SDK** - Package for embedding Activepieces UI in external Angular applications

Architecture Overview



Key Architecture Decisions

Decision	Rationale
Controllers/Services in Core	TypeScript `rootDir` constraint prevents importing from outside package
Hook-based Extension	Core defines hook factories with defaults; BMP provides custom implementations
Entity Protection via .gitattributes	Entities cannot move outside server `rootDir`; protected with `merge=ours`
Frontend Conditional Loading	`isBmpEnabled()` and `getBmpDefaultRoute()` utilities for route/navigation control

Organization Hierarchy

```
Platform
  └── Organization (e.g., "ABC")
        ├── Environment: Dev
        │     ├── Admin User (project owner)
        │     ├── metadata: { ADA_BMP_API_URL: "..." }
        │     └── Project
        ├── Environment: Staging
        │     └── ...
        └── Environment: Production
              └── ...
```

Role Hierarchy

```
SUPER_ADMIN (cross-platform)
  └── OWNER (tenant owner, per platform)
        └── ADMIN (environment admin, has personal project)
              ├── OPERATOR
              └── MEMBER
```

Features Implemented

Backend Features

Feature	API Endpoints	Status
Organization CRUD	`GET/POST /v1/organizations`	✅ Working
Organization Environments	`GET/POST /v1/organization-environments`	✅ Working
Super Admin Management	`GET/POST /v1/super-admin/*`	✅ Working
Account Switching (SUPER_ADMIN→OWNER)	`POST /v1/super-admin/tenants/:id/switch`	✅ Working
Account Switching (OWNER→ADMIN)	`POST /v1/platforms/switch-to-admin/:id`	✅ Working
BMP Piece Environment Metadata	Connection hooks	✅ Working

Frontend Features

Feature	Route/Component	Status
Super Admin Dashboard	`/platform/super-admin`	✅ Working
Owner Dashboard	`/platform/owner-dashboard`	✅ Working
Organizations Management	`/platform/organizations`	✅ Working
Conditional Sidebar Items	Platform sidebar	✅ Working
Role-based Default Routes	`default-route.tsx`	✅ Working
Account Switch Back Button	`switch-back-button.tsx`	✅ Working

File Organization

Extension Package Structure

```
packages/extensions/bmp/
├── src/
│   ├── index.ts                    # Main entry
│   ├── server/
│   │   ├── index.ts               # Server entry
│   │   ├── bmp.module.ts          # Fastify plugin
│   │   ├── controllers/
│   │   │   ├── organization.controller.ts
│   │   │   ├── organization-environment.controller.ts
│   │   │   ├── super-admin.controller.ts
│   │   │   └── account-switching.controller.ts
│   │   ├── services/
│   │   │   ├── organization.service.ts
│   │   │   ├── organization-environment.service.ts
│   │   │   ├── super-admin.service.ts
│   │   │   ├── account-switching.service.ts
│   │   │   └── multi-tenant-auth.service.ts
│   │   └── hooks/
│   │       ├── auth.hooks.ts
│   │       ├── connection.hooks.ts
│   │       ├── engine.hooks.ts
│   │       └── index.ts
│   ├── web/
│   │   ├── index.ts               # Web entry
│   │   ├── routes/
│   │   │   ├── organizations/
│   │   │   ├── super-admin/
│   │   │   └── owner-dashboard/
│   │   ├── components/
│   │   │   ├── super-admin-layout.tsx
│   │   │   ├── switch-back-button.tsx
│   │   │   ├── ada-bmp-environment-select.tsx
│   │   │   └── sidebar/
│   │   ├── api/
│   │   │   ├── organization-api.ts
│   │   │   └── super-admin-api.ts
│   │   └── hooks/
│   │       ├── organization-hooks.ts
│   │       └── super-admin-hooks.ts
│   ├── shared/
│   │   └── organization/
│   │       ├── organization.ts
│   │       └── organization.request.ts
│   └── hooks/
│       ├── types.ts
│       ├── registry.ts
│       └── index.ts
└── package.json

packages/extensions/react-ui-sdk/
├── src/
│   ├── angular/                  # Angular wrappers
│   ├── react/                    # React component wrappers
│   ├── providers/
│   ├── stubs/
│   ├── types/
│   ├── utils/
│   └── index.ts
├── scripts/
├── package.json
```

```
├── BUILD_GUIDE.md
├── BMP_OVERRIDES.md
└── README.md
```

Entities (Protected in Core)

Entity	Location	Prot
`OrganizationEntity`	`packages/server/api/src/app/organization/organization.entity.ts`	`.gitattribu`
`OrganizationEnvironmentEntity`	`packages/server/api/src/app/organization/organization-environment.entity.ts`	`.gitattribu`
`AccountSwitchingActivityEntity`	`packages/server/api/src/app/account-switching/account-switching-activity.entity.ts`	`.gitattribu`

Code Change Analysis

New Files (Low Merge Conflict Risk)

Server - Organization Module (6 files)

- `packages/server/api/src/app/organization/organization.entity.ts`
- `packages/server/api/src/app/organization/organization.service.ts`
- `packages/server/api/src/app/organization/organization.controller.ts`
- `packages/server/api/src/app/organization/organization.module.ts`
- `packages/server/api/src/app/organization/organization-environment.entity.ts`
- `packages/server/api/src/app/organization/organization-environment.service.ts`

Server - Super Admin Module (2 files)

- `packages/server/api/src/app/super-admin/super-admin.controller.ts`
- `packages/server/api/src/app/super-admin/super-admin.module.ts`

Server - Account Switching (2 files)

- `packages/server/api/src/app/account-switching/account-switching-activity.entity.ts`
- `packages/server/api/src/app/account-switching/account-switching-activity.service.ts`

Server - Database Migrations (7 files)

- `1768457416000-AddAccountSwitchingActivity.ts`
- `1769126400000-AddOrganizationTables.ts`
- `1769127000000-AddOrganizationToUserInvitation.ts`
- `1769127500000-AddMetadataToOrganizationEnvironment.ts`
- `1769127600000-AddProjectIdToOrganization.ts`
- `1769127700000-AddMissingOrganizationEnvironments.ts`
- `1771241733000-AddClientIdToUser.ts`

Shared Types (3 files)

- `packages/shared/src/lib/organization/index.ts`
- `packages/shared/src/lib/organization/organization.ts`
- `packages/shared/src/lib/organization/organization.request.ts`

Web - New Pages/Components (15 files)

- `packages/web/src/app/routes/platform/organizations/index.tsx`
- `packages/web/src/app/routes/platform/organizations/environment-metadata-dialog.tsx`
- `packages/web/src/app/routes/platform/organizations/organization-environments-section.tsx`
- `packages/web/src/app/routes/platform/super-admin/index.tsx`
- `packages/web/src/app/routes/platform/super-admin/create-tenant-dialog.tsx`

- `packages/web/src/app/routes/platform/owner-dashboard/index.tsx`
- `packages/web/src/app/components/super-admin-layout.tsx`
- `packages/web/src/app/components/switch-back-button.tsx`
- `packages/web/src/app/components/sidebar/super-admin/index.tsx`
- `packages/web/src/app/connections/ada-bmp-environment-select.tsx`
- `packages/web/src/app/guards/platform-default-route.tsx`
- `packages/web/src/features/platform-admin/api/organization-api.ts`
- `packages/web/src/features/platform-admin/api/organization-hooks.ts`
- `packages/web/src/hooks/super-admin-hooks.ts`
- `packages/web/src/lib/super-admin-api.ts`

Custom Piece

- `packages/pieces/custom/ada-bmp/*` - All files are new

Core Modifications (High Conflict Risk)

Critical Server Files

File	Change Type	BMP-Specific Logic
`app-connection-service.ts`	**Heavy**	`isBmpPiece()` helper, environment metadata fetch
`piece-metadata-controller.ts`	**Heavy**	`organizationEnvironmentMetadata` injection
`user-service.ts`	**Heavy**	Role-based filtering, organization meta info
`database-connection.ts`	**Medium**	Added BMP entities
`app.ts`	**Light**	Added `organizationModule`, `superAdminModule`
`user-entity.ts`	**Light**	Added `organizationId`, `clientId` columns

Critical Shared Files

File	Change Type	BMP-Specific Logic
`user.ts`	**Medium**	Added `SUPER_ADMIN`, `OWNER` enum values
`engine-operation.ts`	**Light**	Added `environmentMetadata` properties

Critical Web Files

File	Change Type	BMP-Specific Logic
`authentication-session.ts`	**Heavy**	Account switching stack
`platform-routes.tsx`	**Medium**	Added BMP routes
`invite-user-dialog.tsx`	**Heavy**	Organization selection
`sidebar-user.tsx`	**Medium**	Switch back button

Core Files Modified Summary (18 files)

Server Package (9 files):

1. `packages/server/api/src/app/app.ts` - Conditional BMP module loading
2. `packages/server/api/src/app/app-connection/app-connection-service/app-connection-service.ts` - Hook integration
3. `packages/server/api/src/app/app-connection/app-connection.controller.ts` - isBmpPiece check
4. `packages/server/api/src/app/authentication/authentication-utils.ts` - authHooks usage
5. `packages/server/api/src/app/core/security/v2/authz/authorize.ts` - authHooks usage
6. `packages/server/api/src/app/helper/system-validator.ts` - BMP prop validators
7. `packages/server/api/src/app/helper/system/system-props.ts` - BMP_ENABLED props

8. `packages/server/api/src/app/trigger/app-event-routing/app-event-routing.module.ts` - Type fix
9. `packages/server/worker/src/lib/execute/jobs/execute-property.ts` - organizationEnvironmentMetadata passing

Web Package (9 files):

1. `packages/web/src/app/components/sidebar/platform/index.tsx` - Conditional BMP items
2. `packages/web/src/app/components/sidebar/super-admin/index.tsx` - BMP check
3. `packages/web/src/app/components/super-admin-layout.tsx` - Suspense wrapper
4. `packages/web/src/app/guards/default-route.tsx` - BMP routing logic
5. `packages/web/src/app/guards/platform-default-route.tsx` - BMP route integration
6. `packages/web/src/app/routes/platform-routes.tsx` - Route filtering
7. `packages/web/src/components/icons/settings2.tsx` - SVG fix
8. `packages/web/src/components/providers/socket-provider.tsx` - Strict Mode fix
9. `packages/web/vite.config.mts` - VITE_BMP_ENABLED env loading

New Core Files Created (3 files)

1. `packages/server/api/src/app/app-connection/connection-hooks.ts` - Hook factory
2. `packages/server/api/src/app/authentication/auth-hooks.ts` - Hook factory
3. `packages/web/src/app/routes/bmp-routes.ts` - BMP route utilities

Implementation Details

Hook Pattern Implementation

Core Hook Factory (auth-hooks.ts):

```
import { hooksFactory } from '../helper/hooks-factory'

export interface AuthHooks {
    isPrivilegedRole: (role: string) => boolean
    getDefaultRoute: (role: string) => string
    enrichToken: (user: User, token: AccessToken) => Promise<AccessToken>
}

export const authHooks = hooksFactory.create<AuthHooks>(_log => ({
    isPrivilegedRole: () => false,  // No privileged roles in community
    getDefaultRoute: () => '/flows',
    enrichToken: async (_, token) => token,
}))
```

BMP Hook Implementation (bmp/hooks/auth.hooks.ts):

```
export const bmpAuthHooks = (log: FastifyBaseLogger): AuthHooks => ({
    isPrivilegedRole: (role: string) => ['SUPER_ADMIN', 'OWNER'].includes(role),
    getDefaultRoute: (role: string) => {
        if (role === 'SUPER_ADMIN') return '/platform/super-admin'
        if (role === 'OWNER') return '/platform/owner-dashboard'
        return '/flows'
    },
    enrichToken: async (user, token) => {
        if (user.organizationId) {
            token.organizationId = user.organizationId
        }
        return token
    },
})
```

Conditional Module Loading (app.ts)

```
import { system, AppSystemProp } from '@activepieces/server-shared'

const bmpEnabled = system.getBoolean(AppSystemProp.BMP_ENABLED) ?? false
if (bmpEnabled) {
    const { bmpModule, bmpHooks } = await import('@activepieces/ext-bmp/server')
    await app.register(bmpModule)

    authHooks.set(bmpHooks.auth)
    connectionHooks.set(bmpHooks.connection)
    projectHooks.set(bmpHooks.project)
}
```

Environment Metadata Flow

```
Organization → OrganizationEnvironment → metadata.ADA_BMP_API_URL
  ↓
Connection validation (app-connection-service.ts)
  ↓
Engine execution (piece-helper.ts injects to process.env)
  ↓
BMP Piece reads process.env.ADA_BMP_API_URL
```

Validation Status

Backend Server

Feature	Status	Notes
Server startup with AP_BMP_ENABLED=true	✅ Validated	BMP modules registered correctly
Server startup with AP_BMP_ENABLED=false	✅ Validated	BMP modules skipped
Organization APIs (/v1/organizations)	✅ Validated	CRUD operations working
Organization Environments APIs	✅ Validated	Initialize, list, update metadata
Super Admin APIs (/v1/super-admin)	✅ Validated	Platform/tenant management
Account Switching (SUPER_ADMIN to OWNER)	✅ Validated	Token exchange working
Account Switching (OWNER to ADMIN)	✅ Validated	Platform switch working
BMP Piece Options (dropdown loading)	✅ Validated	Fixed organizationEnvironmentMetadata passing
App Connections with environment metadata	✅ Validated	ADA_BMP_API_URL correctly injected
Auth Hooks (isPrivilegedRole)	✅ Validated	SUPER_ADMIN, OWNER, ADMIN checks
Connection Hooks (isBmpPiece)	✅ Validated	BMP piece detection working

Frontend Web

Feature	Status	Notes
SUPER_ADMIN default route (/super-admin)	✅ Validated	BMP route working
OWNER default route (/owner-dashboard)	✅ Validated	BMP route working
ADMIN/MEMBER default route (/projects/.../flows)	✅ Validated	Fixed to use project flows

Feature	Status	Notes
Platform sidebar (BMP items conditional)	✅ Validated	Items hidden when BMP disabled
Super Admin sidebar	✅ Validated	Shows for privileged roles
Navigation loop prevention	✅ Validated	Fixed Suspense boundaries
WebSocket reconnection	✅ Validated	Fixed Strict Mode handling
SVG icon rendering (cx attribute)	✅ Validated	Fixed motion.circle initial state
Loading states (Suspense)	✅ Validated	LoadingScreen shown during data fetch

React UI SDK

Feature	Status	Notes
SDK bundle build	✅ Validated	12.9 MB bundle created
SDK package linking	✅ Validated	Linked to bmp-fe-web
Angular app compilation	✅ Validated	No errors
SDK script serving (/sdk/index.js)	✅ Validated	Available via proxy
API proxy configuration	✅ Validated	/api to localhost:3000
Auto-provision endpoint	✅ Validated	Token exchange working

Custom Piece (ada-bmp)

Feature	Status	Notes
Piece loading (AP_DEV_PIECES)	✅ Validated	Piece appears in UI
Environment metadata injection	✅ Validated	ADA_BMP_API_URL available
Channel dropdown options	✅ Validated	Fixed worker metadata passing

Known Non-Critical Warnings

Warning	Status	Notes
getNodesBounds deprecation	⚠️ Not Fixed	React Flow warning, not blocking
path.extname browser warning	⚠️ Not Fixed	Third-party dependency, harmless

Configuration

Environment Variables

Variable	Type	Default	Description
`AP_BMP_ENABLED`	boolean	`false`	Enable/disable BMP extension
`AP_BMP_ORGANIZATIONS`	boolean	`true`	Enable organization features
`AP_BMP_SUPER_ADMIN`	boolean	`true`	Enable super admin features
`AP_BMP_ACCOUNT_SWITCHING`	boolean	`true`	Enable account switching
`AP_DEV_PIECES`	string	-	Set to `ada-bmp` for local development

System Props (server-shared)

```
export enum AppSystemProp {
    BMP_ENABLED = 'AP_BMP_ENABLED',
    BMP_ORGANIZATIONS_ENABLED = 'AP_BMP_ORGANIZATIONS',
    BMP_SUPER_ADMIN_ENABLED = 'AP_BMP_SUPER_ADMIN',
    BMP_ACCOUNT_SWITCHING_ENABLED = 'AP_BMP_ACCOUNT_SWITCHING',
}
```

Vite Configuration (packages/web/vite.config.mts)

```
define: {
    'import.meta.env.VITE_BMP_ENABLED': JSON.stringify(process.env.AP_BMP_ENABLED ?? 'false'),
}
```

tsconfig.base.json Path Aliases

```
{
  "paths": {
    "@activepieces/ext-bmp": ["packages/extensions/bmp/src/index.ts"],
    "@activepieces/ext-bmp/*": ["packages/extensions/bmp/src/*"],
    "@activepieces/react-ui-sdk": ["packages/extensions/react-ui-sdk/src/index.ts"],
    "@activepieces/piece-ada-bmp": ["packages/pieces/custom/ada-bmp/src/index.ts"]
  }
}
```

.gitattributes Merge Protection

```
# BMP Extension Files - Protect from upstream merges
packages/extensions/** merge=ours
packages/pieces/custom/** merge=ours
packages/server/api/src/app/organization/** merge=ours
packages/server/api/src/app/super-admin/** merge=ours
packages/server/api/src/app/account-switching/** merge=ours
```

React UI SDK

Purpose

The React UI SDK enables embedding Activepieces UI components in external applications (specifically the Angular-based bmp-fe-web application).

Key Features

- **CE (Community Edition) only** - Explicitly excludes EE features
- **MIT licensed** for external distribution
- Wraps `packages/web` React components for Angular consumption
- Provides `data-ap-view` attributes for CSS scoping in host apps
- Supports Flow Builder, Dashboard, Connections, Runs, Templates views

Building the SDK

```
npx nx run react-ui-sdk:bundle
```

Integration with bmp-fe-web

In `bmp-fe-web/package.json`:

```
{
  "dependencies": {
    "@activepieces/react-ui-sdk": "file:../../POC/activepieces/dist/packages/react-ui-sdk-bundled",
    "@activepieces/shared": "file:../../POC/activepieces/packages/shared"
  }
}
```

Lessons Learned

TypeScript rootDir Constraint

Issue: Entity migration to `packages/extensions/bmp/src/server/entities/` failed.

Root Cause: The server package's TypeScript configuration (`packages/server/api/tsconfig.app.json`) sets `rootDir: "."` which prevents importing TypeScript files from outside the package directory during compilation.

Error:

```
TS2307: Cannot find module '@activepieces/ext-bmp/entities'
```

Resolution: Keep entities in core with `.gitattributes` protection. Entities are just schema definitions with no business logic.

React Suspense with useSuspenseQuery

Issue: "Maximum update depth exceeded" navigation loops.

Root Cause: `useSuspenseQuery` suspends the component (doesn't return `isLoading`), causing routing decisions before data was available.

Resolution: Wrap components using `useSuspenseQuery` with `React.Suspense` and `LoadingScreen` fallback.

Zod/TypeBox Validator Mismatch

Issue: API validation errors ("Cannot read properties of undefined (reading 'run')").

Root Cause: Global Zod `validatorCompiler` was applied to controllers still using TypeBox schemas.

Resolution: Convert affected controllers from `FastifyPluginAsyncTypebox` to `FastifyPluginAsyncZod` with Zod schemas.

WebSocket Strict Mode

Issue: WebSocket connections failing immediately.

Root Cause: React Strict Mode double-invokes effects, causing duplicate socket connections and premature cleanup.

Resolution: Track connection count, use refs for cleanup state, add reconnection logic.

Maintenance Guide

Upgrading Upstream Activepieces

1. **Fetch upstream changes:**

```
git fetch upstream
git merge upstream/main
```

2. **Review conflicts:** `.gitattributes` with `merge=ours` will preserve BMP files automatically.
3. **Manual review needed for:**
   - Changes to hook factory interfaces in core
   - New system props that might conflict
   - Database migration ordering
4. **Re-validate after merge:**

```
./scripts/dev-local.sh start
# Test with AP_BMP_ENABLED=true and AP_BMP_ENABLED=false
```

Adding New BMP Features

1. **Server-side:** Add services/controllers to `packages/server/api/src/app/` (protected by `.gitattributes`)
2. **Web-side:** Add routes/components to `packages/web/src/app/routes/platform/`
3. **Update hooks** if new extension points needed
4. **Add migrations** to `packages/server/api/src/app/database/migration/postgres/`

Testing Checklist

- ☐ Server starts with `AP_BMP_ENABLED=true`
- ☐ Server starts with `AP_BMP_ENABLED=false`
- ☐ BMP routes respond correctly when enabled
- ☐ Core routes work when BMP disabled
- ☐ TypeScript compilation passes
- ☐ No circular dependencies
- ☐ SDK bundle builds successfully
- ☐ bmp-fe-web integration works

Appendix: API Reference

Organization APIs

Method	Endpoint	Description
GET	`/v1/organizations`	List organizations
POST	`/v1/organizations`	Create organization
GET	`/v1/organizations/:id`	Get organization
POST	`/v1/organizations/check-admin`	Check admin availability

Super Admin APIs

Method	Endpoint	Description
GET	`/v1/super-admin/platforms`	List platforms
GET	`/v1/super-admin/tenants`	List tenants
POST	`/v1/super-admin/tenants`	Create tenant
GET	`/v1/super-admin/stats`	Get statistics
POST	`/v1/super-admin/tenants/:platformId/switch`	Switch to tenant

Account Switching APIs

Method	Endpoint	Description
POST	`/v1/platforms/switch-to-admin/:adminId`	OWNER switches to ADMIN

Document generated from BMP Extension implementation project, March 2026